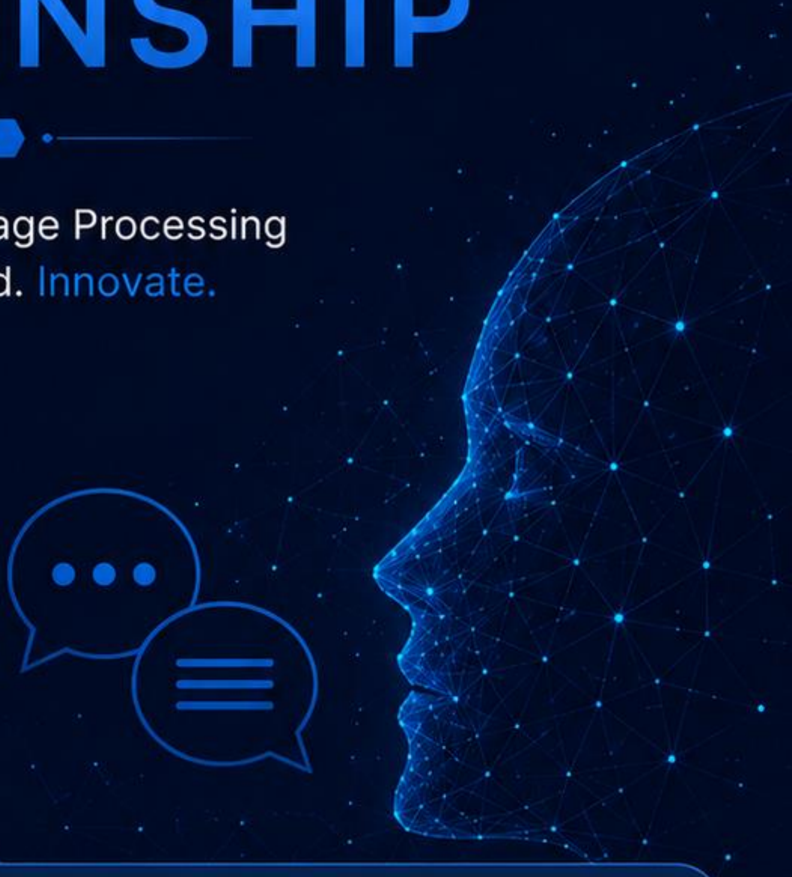
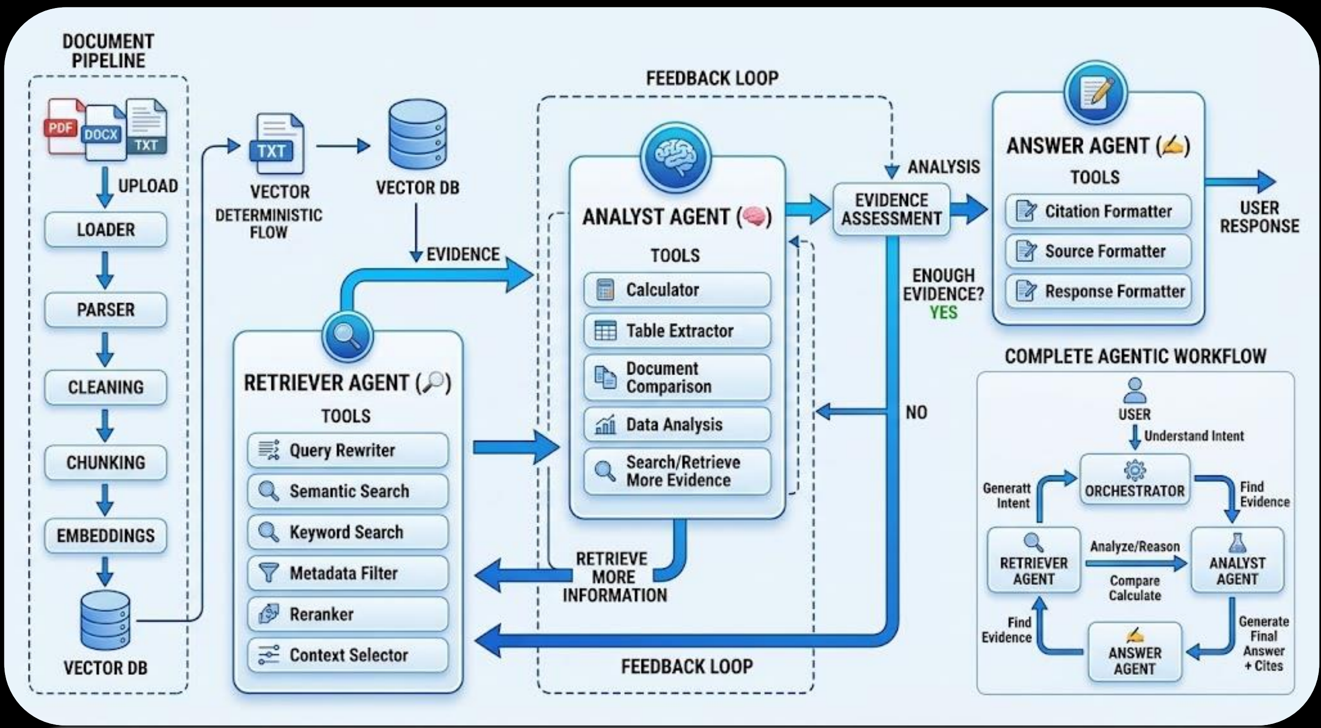
Cellula Technologies

NLP INTERNSHIP

Natural Language Processing
Learn. Build. Innovate.



Project 3



DOCUMENT PIPELINE
PDF
DOCX
TXT
UPLOAD
LOADER
PARSER
CLEANING
CHUNKING
EMBEDDINGS
VECTOR DB
TXT
VECTOR
DETERMINISTIC FLOW
VECTOR DB
EVIDENCE
FEEDBACK LOOP
RETRIEVER AGENT (🔎)
TOOLS
Query Rewriter
Semantic Search
Keyword Search
Metadata Filter
Reranker
Context Selector
ANALYST AGENT (🧠)
TOOLS
Calculator
Table Extractor
Document Comparison
Data Analysis
Search/Retrieve More Evidence
RETRIEVE MORE INFORMATION
FEEDBACK LOOP
ANALYSIS
EVIDENCE ASSESSMENT
ENOUGH EVIDENCE?
YES
NO
ANSWER AGENT (✍)
TOOLS
Citation Formatter
Source Formatter
Response Formatter
USER RESPONSE
COMPLETE AGENTIC WORKFLOW
USER
Understand Intent
ORCHESTRATOR
Generatt Intent
Find Evidence
RETRIEVER AGENT
Analyze/Reason
Compare Calculate
ANALYST AGENT
Find Evidence
ANSWER AGENT
Generate Final Answer + Cites

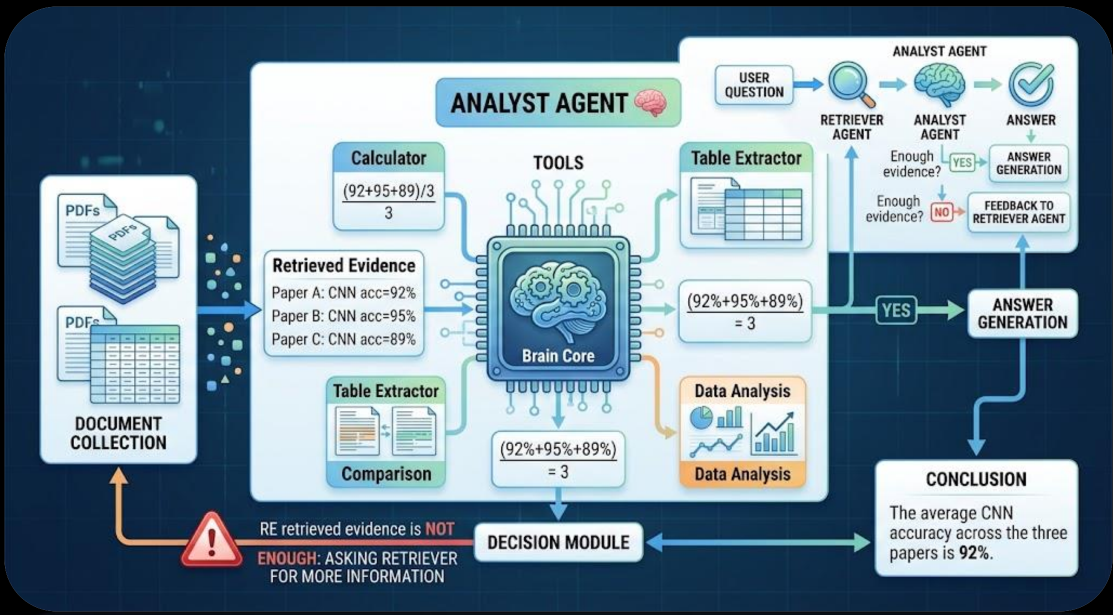
This system is designed to help you chat with your documents using three specialized AI agents working together as a team. Instead of relying on a single AI to handle everything, the workload is divided into specific roles to give you accurate, well-reasoned answers backed by exact sources.

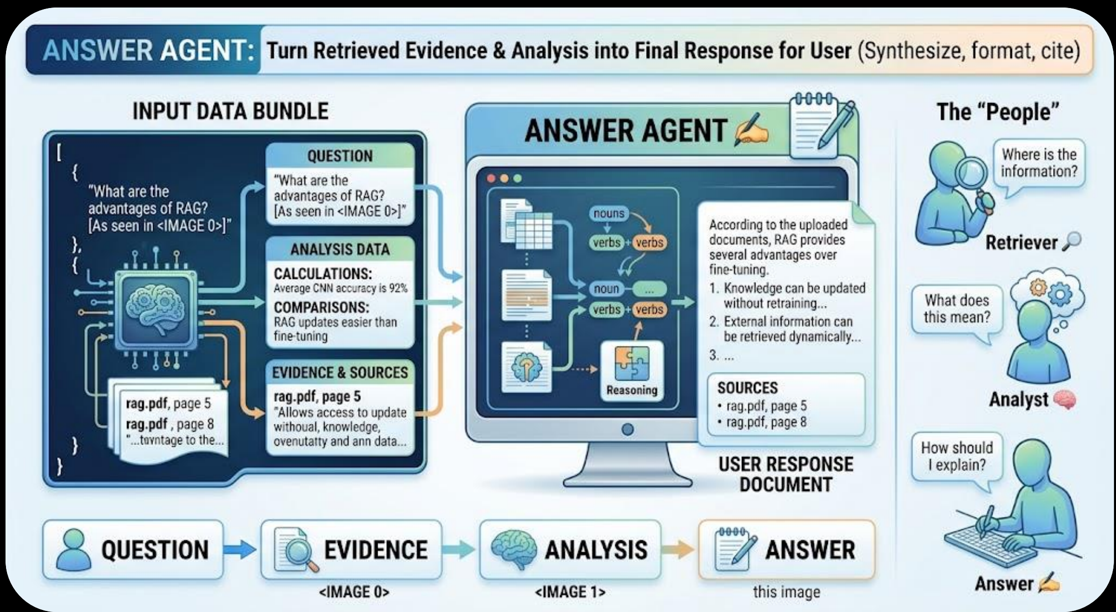
1. Document Preparation Pipeline Before the agents start working, your uploaded files (PDFs, Word documents, or text files) go through an automatic setup process. The system cleans up the text, breaks it into smaller, readable chunks, converts those chunks into searchable numerical data (embeddings), and stores them in a Vector Database. This step makes searching through long documents fast and precise.

2. The 3-Agent Workflow When you ask a question, an **Orchestrator** coordinates the process across three dedicated agents:

- **Retriever Agent (🔎):** Its sole job is to search the Vector Database and find relevant excerpts (evidence). It uses tools like query rewriting, semantic search, keyword matching, and reranking to locate the right information without attempting to answer the question itself.

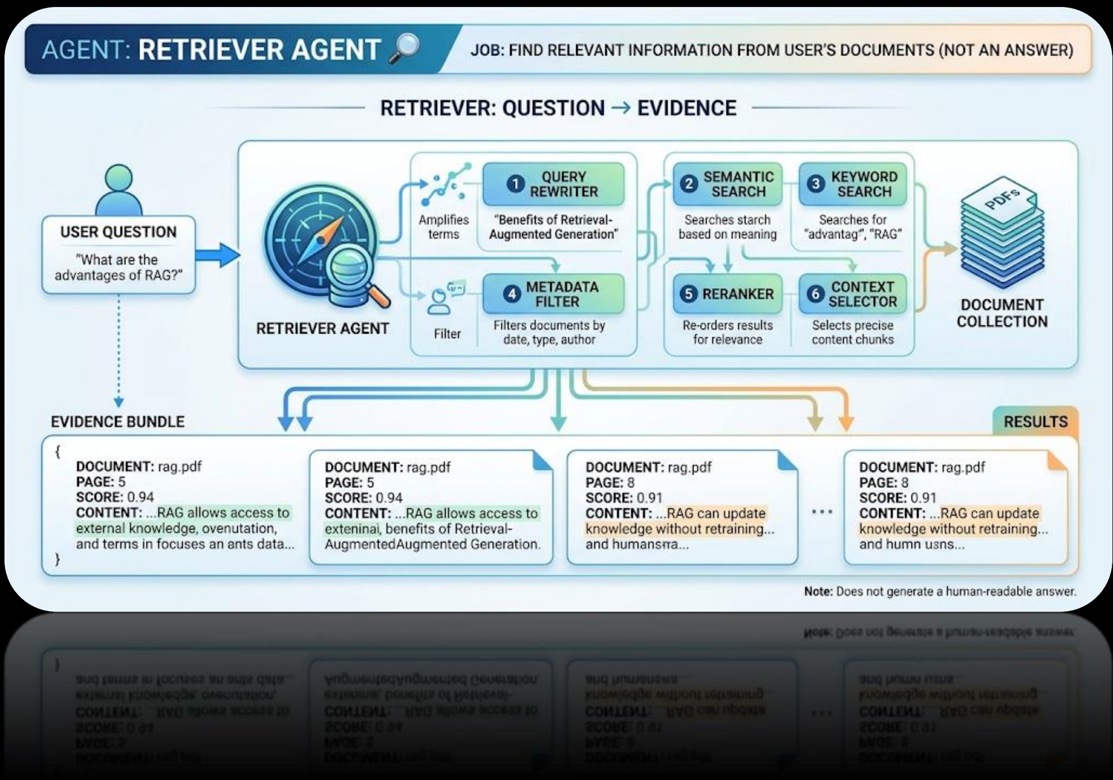
- **Analyst Agent (🧠):** This agent takes the evidence gathered by the Retriever and performs the heavy lifting. It evaluates the data, compares facts, extracts tables, and performs calculations. Most importantly, if the Analyst notices missing information, it triggers a **Feedback Loop** back to the Retriever Agent to fetch additional details before moving forward.
- **Answer Agent (✍️):** Once the Analyst determines there is enough solid evidence, the Answer Agent crafts the final, easy-to-read response. It organizes the insights into clear text or tables and uses formatting tools to attach precise page and document citations.


ANALYST AGENT
USER QUESTION
RETRIEVER AGENT
ANALYST AGENT
ANSWER
Enough evidence?
YES
ANSWER GENERATION
Enough evidence?
NO
FEEDBACK TO RETRIEVER AGENT
ANALYST AGENT
Calculator
(92+95+89)/3
3
TOOLS
Table Extractor
PDFs
PDFs
PDFs
DOCUMENT COLLECTION
Retrieved Evidence
Paper A: CNN acc=92%
Paper B: CNN acc=95%
Paper C: CNN acc=89%
Brain Core
(92%+95%+89%)
= 3
YES
ANSWER GENERATION
Table Extractor
Comparison
(92%+95%+89%)
= 3
Data Analysis
Data Analysis
CONCLUSION
The average CNN accuracy across the three papers is 92%.
RE retrieved evidence is NOT
ENOUGH: ASKING RETRIEVER FOR MORE INFORMATION
DECISION MODULE


ANSWER AGENT: Turn Retrieved Evidence & Analysis into Final Response for User (Synthesize, format, cite)
INPUT DATA BUNDLE
"What are the advantages of RAG? [As seen in <IMAGE 0>]"
QUESTION
"What are the advantages of RAG? [As seen in <IMAGE 0>]"
ANALYSIS DATA
CALCULATIONS:
Average CNN accuracy is 92%
COMPARISONS:
RAG updates easier than fine-tuning
EVIDENCE & SOURCES
rag.pdf, page 5
"Allows access to update withoual, knowledge, ovenutatty and ann data...
rag.pdf, page 5
rag.pdf, page 8
"...tavntage to the...
ANSWER AGENT
nouns
verbs
verbs
noun
verbs
verbs
Reasoning
According to the uploaded documents, RAG provides several advantages over fine-tuning.
1. Knowledge can be updated without retraining...
2. External information can be retrieved dynamically...
3. ...
SOURCES
• rag.pdf, page 5
• rag.pdf, page 8
USER RESPONSE DOCUMENT
The "People"
Where is the information?
Retriever
What does this mean?
Analyst
How should I explain?
Answer
QUESTION
EVIDENCE
ANALYSIS
ANSWER
<IMAGE 0>
<IMAGE 1>
this image

Retriever Agent

Find the relevant information from the user's documents.



Tools :

1. Query Rewriter

The **Query Rewriter** transforms the user’s original question into a clearer and more retrieval-friendly query. Users often ask questions using vague language, pronouns, incomplete sentences, or conversational expressions. For example, if the previous question was “What is RAG?” and the user asks “What are its disadvantages?”, the Query Rewriter can transform it into “What are the disadvantages of Retrieval-Augmented Generation (RAG)?” It can also expand abbreviations, clarify terminology, and generate alternative queries when necessary. The rewritten query is then passed to the retrieval system. Its main purpose is to improve the quality and relevance of the documents retrieved from the knowledge base.

2. Semantic Search

Semantic Search retrieves document chunks based on their meaning rather than simply matching exact words. First, the user's query is converted into an embedding vector using an embedding model. Document chunks were previously converted into vectors and stored in a vector database. The system compares the query vector with stored document vectors using similarity measures such as cosine similarity. Chunks with the highest similarity scores are considered the most semantically relevant. This allows the system to retrieve useful information even when the user and document use different words with similar meanings. Semantic Search is therefore a fundamental component of modern Retrieval-Augmented Generation systems.

3. Keyword Search

Keyword Search retrieves documents by looking for specific words or phrases appearing in the query and documents. Unlike Semantic Search, it focuses primarily on lexical or exact-term matching. This makes it particularly useful when the user searches for specific technical terms, names, equations, IDs, or unique phrases. For example, searching for “BERT-base” or “Equation 12” may benefit greatly from keyword-based retrieval. Traditional approaches such as inverted indexes and BM25 can be used to rank matching documents. Keyword Search can also complement Semantic Search because semantic retrieval may sometimes miss important exact terms. Combining both approaches creates a more robust retrieval system.

4. Metadata Filter

The **Metadata Filter** restricts document retrieval according to additional information associated with each document chunk. During document ingestion, metadata such as document name, page number, chapter, section, author, date, or document type can be stored alongside the text. When a user asks a specific question, the filter can limit the search to relevant metadata. For example, if the user says "According to Chapter 4 of paper.pdf," the system can search only chunks belonging to that chapter and document. This reduces irrelevant results, improves retrieval accuracy, and can significantly reduce the amount of information that subsequent agents need to process.

5. Reranker

The **Reranker** improves the ordering of results returned by the initial retrieval process. Semantic or hybrid search may retrieve a relatively large number of candidate chunks, such as the top 20 or 50 results. A more sophisticated ranking model then examines the relationship between the query and each retrieved chunk and assigns a new relevance score. The results are subsequently reordered according to these scores, and only the best chunks may be passed to the Analyst Agent.

This creates a two-stage retrieval process: fast initial retrieval followed by more accurate reranking. Reranking is particularly useful when the document collection is large or contains highly similar information.

6. Context Selector

The **Context Selector** determines which retrieved chunks should ultimately be provided to the Analyst Agent. Even after reranking, several retrieved chunks may contain redundant, irrelevant, or excessively similar information. The Context Selector chooses a smaller set of high-quality evidence while considering relevance, diversity, document coverage, and available context length. For example, instead of passing 20 retrieved chunks to the Analyst, it may select the best 5–8 chunks. It can also preserve important metadata such as document names and page numbers for later citation. Its main purpose is to create a compact, informative context that gives the Analyst enough evidence without unnecessarily increasing the model's context size.

Analyst Agent

Understand, combine, compare, calculate, and reason over the retrieved information**.**

1. Calculator

The **Calculator** tool allows the Analyst Agent to perform accurate numerical calculations instead of relying on the language model to calculate values internally. It can handle operations such as addition, subtraction, multiplication, division, percentages, averages, ratios, and statistical calculations. For example, if several documents report different accuracy values, the Analyst can use the Calculator to calculate their average accurately. It can also calculate differences between models, percentage improvements, and other numerical comparisons. This reduces arithmetic errors and makes the analysis more reliable. The Calculator is especially useful when documents contain experimental results, financial values, measurements, statistics, or numerical performance metrics.

2. Table Extractor

The **Table Extractor** tool is responsible for identifying and extracting structured tables from documents. Tables in PDFs often contain important information such as accuracy, precision, recall, F1-score, model parameters, or experimental results. Instead of treating the table as ordinary text, the tool converts it into a structured representation that the Analyst Agent can easily process. For example, a table containing model names and their performance metrics can be transformed into rows and columns. The Analyst can then compare values, calculate statistics, and identify the best-performing model. This tool is particularly important when the documents contain scientific papers, reports, financial documents, or datasets.

3. Document Comparison

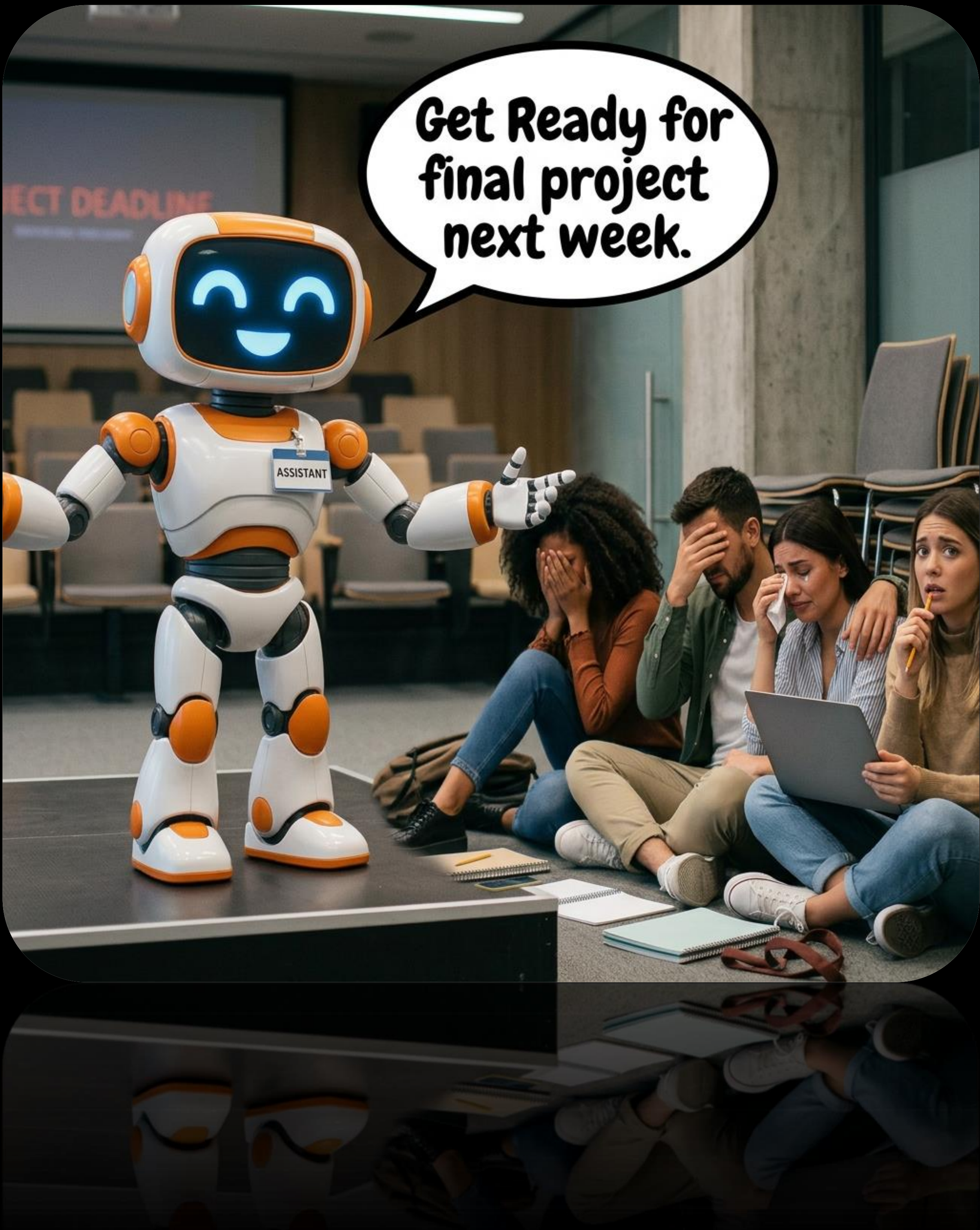
The **Document Comparison** tool helps the Analyst Agent compare information across multiple documents. It can identify similarities, differences, advantages, disadvantages, methodologies, results, and conclusions presented in different sources. For example, if the user uploads three research papers and asks which model performs best, the tool can organize the relevant information from each paper into a comparable structure. The Analyst can then reason over the differences and produce a meaningful comparison. This tool is particularly useful for questions involving multiple documents, such as comparing research papers, technical reports, product specifications, methodologies, experimental results, or competing approaches described across different sources.

4. Data Analysis

The **Data Analysis** tool allows the Analyst Agent to perform structured analysis on numerical or tabular information extracted from documents. It can calculate averages, percentages, differences, rankings, distributions, and other useful statistics. For example, if several papers provide model accuracy, precision, and recall values, the tool can organize these values and determine which model performs best overall. It can also identify trends and relationships between different variables. This tool becomes especially useful when the user's question requires more than simply retrieving information. Instead, it enables the Analyst Agent to transform raw evidence into meaningful quantitative insights and support conclusions with calculated results.

5. Search / Retrieve More Evidence

The **Search / Retrieve More Evidence** tool allows the Analyst Agent to request additional information when the evidence initially provided by the Retriever Agent is insufficient. Instead of making assumptions or generating unsupported conclusions, the Analyst can send a more specific query back to the Retriever. For example, if the Analyst receives information about model accuracy but needs computational complexity to complete a comparison, it can request evidence specifically related to complexity. This creates a feedback loop between the Analyst and Retriever. The process can continue until sufficient evidence is available, making the system more flexible and reducing the risk of incomplete or unsupported analysis.


Get Ready for final project next week.
ASSISTANT